

Hours

Duration: 3 Hours

Note:

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- Which one of the following robots also called spherical robot?
A) SCARA
B) PUMA
C) Polar
D) Cylindrical
- In which type of kinematic chain are all joints connected in a loop?
A) Closed kinematic chain
B) Open kinematic chain
C) Neither
D) Serial Kinematic Chain
- SCARA robots are often used in which of the following applications?
A) Deep-sea exploration
B) Picking and placing in assembly lines
C) Agricultural drones
D) Search and rescue
- What component in a robotic manipulator is responsible for producing motion?
A) Joints
B) Links
C) Actuators
D) Sensors
- If $a_{xyz} = (4, 3, 2)^T$ are the coordinates with respect to the reference coordinate system, determine the corresponding points a_{uvw} , with respect to the rotated OUVW coordinate system if it has been rotated 60° about the OZ axis.
A) $\begin{bmatrix} -4.598 \\ 1.964 \\ 2 \end{bmatrix}$
B) $\begin{bmatrix} 1.964 \\ 4.598 \\ 3 \end{bmatrix}$
C) $\begin{bmatrix} 1.964 \\ 4.598 \\ -3 \end{bmatrix}$
D) $\begin{bmatrix} 4.598 \\ -1.964 \\ 2 \end{bmatrix}$
- What is the primary mechanism by which vacuum grippers hold objects?
A) Magnetic force
B) Mechanical interlock
C) Adhesive
D) Atmospheric pressure
- Given a point $a_{uvw} = (6, 2, 4)^T$ with respect to the rotated OUVW coordinate system, determine corresponding point a_{xyz} with respect to the reference coordinate system if it has been rotated 60° about OZ axis
A) $\begin{bmatrix} 1.45 \\ 4.964 \\ -2 \end{bmatrix}$
B) $\begin{bmatrix} 0.896 \\ 4.964 \\ 2 \end{bmatrix}$
C) $\begin{bmatrix} 1.268 \\ 6.196 \\ 4 \end{bmatrix}$
D) $\begin{bmatrix} -4 \\ -1.26 \\ -2 \end{bmatrix}$

8. A point $a_{uvw} = (5, 2, 3)$ is to be translated a distance -3 units along the OX axis and 5 units along the OZ axis. Using the appropriate homogeneous transformation matrix, determine the new point a_{xyz} .
- A) $\begin{bmatrix} 2 \\ 2 \\ 8 \\ 1 \end{bmatrix}$ B) $\begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$
- C) $\begin{bmatrix} 2 \\ 3 \\ 0 \\ 1 \end{bmatrix}$ D) $\begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix}$
9. What type of actuator uses fluid to generate motion?
- A) Electric actuator B) Pneumatic actuator
C) Hydraulic actuator D) Solar actuator
10. Which motor often requires a complex control circuit?
- A) DC motor B) Stepper motor
C) AC motor D) Hydraulic motor
11. Which actuator type would generally be safest to use in a flammable environment?
- A) Pneumatic actuator B) Hydraulic actuator
C) Magnetic actuator D) Electric actuator
12. When selecting an electric actuator for a high-torque, low-speed application, which motor would you consider?
- A) Stepper motor B) DC servo motor
C) Linear motor D) AC motor
13. From which of the following is known as contact sensor.
- A) Tactile sensor B) Capacitive Proximity sensor
C) Ultrasonic Sensor D) IR sensor
14. Which physical quantity does an LVDT primarily measure?
- A) Temperature B) Angular velocity
C) Pressure D) Linear displacement
15. What is the typical output of an LVDT when the core is perfectly centered?
- A) Zero voltage B) Negative voltage
C) Maximum voltage D) Variable voltage
16. In an incremental encoder, how rotation is typically determined?
- A) By counting the number of pulses B) By the encoder's serial number
C) By measuring the voltage level D) By measuring the current level
17. What is a key attribute of bio-inspired robotics?
- A) Replicating human speech B) Using solar energy
C) Being inspired by nature D) Ability to cook
18. In which gait do all four legs of the robot leave the ground at the same time?
- A) Walk B) Gallop
C) Trot D) Bound
19. In which application would a pneumatic actuator be least appropriate?
- A) Painting robots B) Packaging machinery
C) Surgical robots D) Food processing
20. Which of the this is a key challenge in implementing safe and efficient autonomous navigation in UAVs?
- A) Improving aerodynamic design B) Real-time obstacle detection and avoidance
C) Color customization D) Lighter materials

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define the resolution, accuracy, and repeatability with respect to robot manipulator. Explain the factors that effect resolution, accuracy, and repeatability.	8	L*2	1	1

- b) Obtain ${}^A T_B$, ${}^B T_C$, ${}^C T_D$ and ${}^D T_A$.

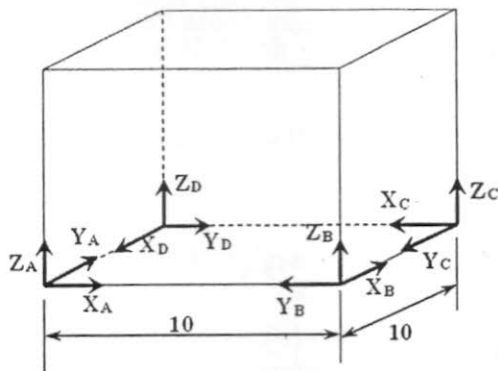


Fig. 1 (b)

8 L3 2 2

2. a) Explain the term "Degrees of Freedom". With a suitable robot manipulator explain the 6 degree of freedom system where 3-degree of freedom is used for body and arm movements and the remaining 3-degree of freedom for wrist movements.
- b) A robotic workstation has been setup with a TV camera as shown in Fig.Q2(b). The camera can see the robot coordinate system where a six joint robot is attached. It can also see the centre of an object (Cube) to be manipulated by the robot. If local coordinate system is established at the centre of the cube, this object as seen by the camera can be represented by a homogeneous transformation matrix ${}^{camera}T_{cube}$. If the origin of the base coordinate system as seen by the camera can also be expressed by the homogeneous transformation matrix ${}^{camera}T_{base}$.

8 L1 1 1

$${}^{camera}T_{cube} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 10 \\ 0 & 0 & -1 & 9 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{camera}T_{base} = \begin{bmatrix} 1 & 0 & 0 & -10 \\ 0 & -1 & 0 & 20 \\ 0 & 0 & -1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Determine the position and orientation of cube coordinate system with respect to the base coordinate system.

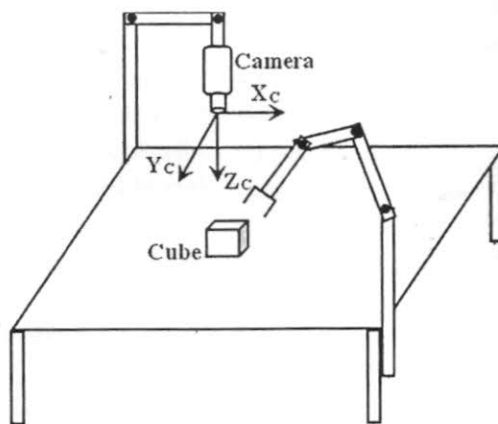


Fig Q. 2(b)

8 L3 2 1

3. a) The telescopic arm of an industrial robot obtains total range of rotation of 100 degree. The robot has a 8bit storage capacity for the axis. The arm fully extends to 1400 mm and fully retracts to 600 mm from the pivot point. Determine the robots control resolution:
- For the axis in degrees of rotation
 - On linear scale in fully extended and retracted position.

6 L2 1 1